Lab 02 - Regression

1. Introduction

This project focuses on predicting song popularity using audio features and metadata. Two approaches were implemented: Submission 1 used Linear Regression as a baseline combined with boosting models, while Submission 2 used an optimized LightGBM pipeline with advanced encoding techniques. Performance was evaluated using RMSE along with MAE, R^2, and MAPE to provide a more comprehensive assessment of model performance.

2. Dataset

The dataset consists of training and test sets with 61 features representing audio characteristics, derived metrics, and metadata. The target variable is a continuous popularity score (0–100).

3. Data preprocessing

Submission 1:

Feature engineering includes extracting year, month, day, and weekday from publication_timestamp, allowing the model to capture temporal patterns such as seasonal trends. Interaction and aggregated features were created to introduce relationships between variables. Missing values were handled using median imputation (robust to outliers) for numerical features and "Unknown" for categorical features. One-hot encoding was applied using a ColumnTransformer.

Submission 2:

A more optimized preprocessing strategy was used. The track_identifier was removed to avoid noise. Year and month features were extracted to capture temporal effects. Frequency encoding was applied to high-cardinality categorical variables to reduce dimensionality compared to one-hot encoding. Ordinal encoding was used for low-cardinality features, while target encoding was applied to high-cardinality features to incorporate label-dependent information. These encoding techniques help the model learn more meaningful patterns while avoiding excessive feature expansion.

4. Approaches Used

In Submission 1, Linear Regression was used as a baseline because it is simple, fast, and provides a reference point for performance. However, it assumes a linear relationship between features and the target, which limits its ability to capture complex patterns in the data. To address this, boosting models such as CatBoost, XGBoost, and LightGBM were used, as they build ensembles of decision trees that can model non-linear relationships and feature interactions.

In Submission 2, LightGBM was selected as the primary model due to its efficiency and strong performance on tabular data. It uses histogram-based learning and leaf-wise tree growth, which allows it to capture complex structures while remaining computationally efficient. However, these models require careful tuning and are less interpretable compared to linear models.

Boosting models work by sequentially training decision trees, where each new tree focuses on correcting the errors made by previous ones. This iterative process improves model accuracy and allows the model to learn complex relationships in the data.

Approach and Models used	**Description**	**Why Chosen**	**Advantages**	**Disadvantages**
Linear Regression (Submission 1 Baseline)	A baseline regression model that predicts popularity as a weighted linear combination of input features after preprocessing.	Used as a fast and interpretable benchmark to compare against more complex models.	Very fast training and inference, simple to interpret, low computational cost.	Cannot capture non-linear relationships or complex feature interactions, leading to lower accuracy on rich datasets.

Advanced Boosting Models (Submission 1: CatBoost, XGBoost, LightGBM, Gradient Boosting + Blending)	Ensemble models that build boosted decision trees; multiple models are compared and optionally blended for better performance.	Chosen to capture non-linear patterns and feature interactions and improve leaderboard performance.	High predictive power, handles tabular data well, strong performance in competitions.	More complex to tune, higher training time, less interpretable, risk of overfitting.
LightGBM Focused Pipeline (Submission 2)	A single optimized LightGBM model trained on engineered and encoded features (frequency, ordinal, target encoding).	Selected for a strong balance between accuracy and efficiency on tabular data.	Fast training, scalable, high accuracy with proper feature engineering.	Sensitive to hyperparameters and preprocessing, less interpretable than simple models.

5. Evaluation Metrics

1. Root Mean Squared Error (RMSE): Main competition metric, Penalizes large errors
2. Mean Absolute Error (MAE): Average absolute difference between predictions and actual values
3. R^2 Score: Measures variance explained by the model
4. Mean Absolute Percentage Error (MAPE): Measures relative error percentage

Lower RMSE and MAE values in Submission 2 indicate improved prediction accuracy and reduced average error. The higher R^2 score (0.78 vs 0.67) shows that the LightGBM model explains more variance in the data. Additionally, the consistency between validation and leaderboard scores suggests better generalization compared to Submission 1. The reduction in MAE alongside RMSE indicates that the model not only reduces large errors but also improves overall prediction consistency. This suggests that the model effectively captures both general trends and individual variations in the data.

6. Results

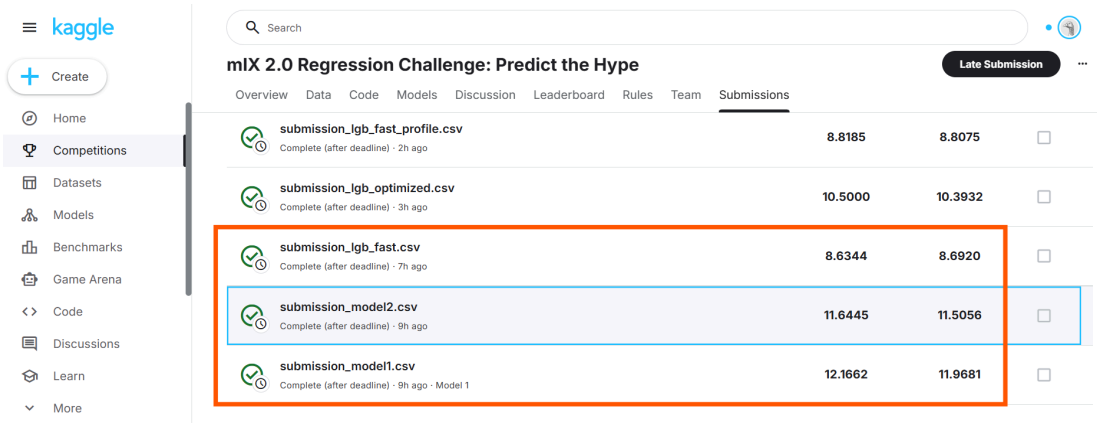
Validation Performance

Submission	Model	RMSE	MAE	R^2	Notes
Submission 1	Linear Regression	12.41	8.18	0.67	Baseline model
Submission 1	CatBoost	11.75	8.55	0.70	Improved over baseline
Submission 1	Blended Model	11.49	-	-	Best validation result in Submission 1
Submission 2	LightGBM	10.08	5.43	0.78	Best overall validation performance

Kaggle Leaderboard Results

Submission	File Name	Private Score (RMSE)	Public Score (RMSE)
Submission 1	submission_model1.csv	12.17	11.97
Submission 1	submission_model2.csv	11.64	11.51
Submission 2	submission_lgb_fast.csv	8.63	8.69

7. Conclusion

Submission 2 performed best, achieving lower RMSE and MAE and a higher R^2, indicating more accurate and reliable predictions. The close match between validation and leaderboard results shows good generalization with minimal overfitting. This improvement is due to LightGBM's ability to model non-linear relationships and feature interactions, along with effective preprocessing. Further gains could be achieved through hyperparameter tuning and improved feature selection.

8. Screenshot of the submission